

PM Narendra Modi, in his Independence Day speech this year, highlighted India's advantages in terms of a trinity of Ds—demography, democracy and diversity

to be 100 times faster than 5G and to handle enormous amount of data transmission without any latency. Its coming will transform the way we communicate at work, home and play. Among its uses will be guiding autonomous cars, enabling 'teleporting' to remote locations in addition to hologram meetings that will combine the physical with the virtual. India could use it for digitising its land records—a major source of disputes as well as corruption—by providing 4D images. It will also vastly improve E-governance, especially the delivery of public services in rural areas while facilitating speedier and secure E-commerce—another major engine of economic growth.

To do all this and more, among the technologies India will have to master is quantum computing, which can process information a thousand times faster than the conventional binary transistors. Such superfast processors will revolutionise the banking, health and defence sectors. India will also need mission-scale effort from both government and industry to acquire leadership in Artificial Intelligence (AI). The prowess of ChatGPT has already captured the popular imagination, but this is just the beginning. It is encouraging that Indian companies are among the world leaders in AI. Mumbai-based Fractal Analytics, for instance, is working with more than 150 Fortune 500 companies.

In addition, India will have to plug into other emerging technologies such as Light Fidelity or LiFi. Unlike WiFi that uses radio frequency, LiFi harnesses the photons emitted by LED to transmit data at 14 times the speed and eliminates the need for electronic devices like routers and modems. Meanwhile, satellite internet facilities provided by Bharti Global, which has a constellation of 634 low earth orbit satellites, could provide high-speed connectivity to remote areas, promising to transform rural India. The country, however, lacks mastery in semiconductor chips and urgently needs to develop indigenous competence. In short, what we need to build in the next two decades is a powerful Indiaverse.

Energy comes next. Currently, India imports as much as 85 per cent of its crude oil requirement and spent Rs 13 lakh crore, or close to 4 per cent of its GDP, in 2022-23 on this. That's unsustainable both in terms of financial costs and given the worldwide